BRAXTON VOGEL

SOFTWARE ENGINEER

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PROFESSIONAL SUMMARY

Software Engineering student at Sam Houston State University (SHSU) with expected graduation date of December 2027, pursuing a future M.S. in & Data Science. Dean's List (Fall 2025, Spring 2026). Strong analytical and problem-solving skills with an applied focus on data analytics, AI-assisted workflows, and technical communication. Proven adaptability through leadership and customer-facing roles.

EDUCATION

Bachelor of Science in Software Engineering; Sam Houston State University (SHSU), Huntsville, Texas

Certification

- Google Data Analytics Professional Certification
  - SQL Skills
  - Data Cleaning & Transformation Skills
  - Data Visualization Skills
  - Spreadsheet & Basic R Skills

Relevant Pre-College Coursework; Klein Cain High School, Klein, Texas

- Completed both General Academic and Video Game Programming Pathways, developing programming skills through structured coursework and applied projects, which laid the foundation for independent software projects

PROJECT *Additional demos, visuals, and technical breakdowns available in portfolio*

SammyOS – Native Desktop AI Workspace | GitHub *(braxtonvogel)* Tauri, Rust, Next.js, TypeScript, Python

- Built a native Windows desktop AI workspace using Tauri + Next.js, integrating real-time screen capture, voice transcription, file/codebase analysis, and autonomous web research into a single unified assistant named Sam
- Designed a multi-provider LLM routing system with automatic fallback across user-supplied keys (OpenAI, Anthropic, Groq, custom endpoints), Ollama (local), Groq, Gemini, and Cerebras — ensuring near-zero downtime
- Implemented a full authentication system with JWT session tokens, per-user encrypted API key storage, and rate-limited auth routes (IP-level and per-account) using Upstash Redis and bcrypt
- Architected a distributed cloud job pipeline via QStash for vault file analysis and autonomous research jobs, with polling, timeout guards, and real-time status updates in the UI
- Developed a live telemetry dashboard (sammyos-live.vercel.app) tracking aggregate events — sessions, messages, research runs, and vault uploads fed by a dedicated nexus-analyzer backend
- Wrote a Python-based silent launcher (SammyOS.vbs + launch.py) with dependency verification, GitHub auto-update checks, Task Scheduler integration, and double-launch prevention

Student Risk Prediction System (AI + ML Web Application) | GitHub *(braxtonvogel)* Python, Streamlift, Scikik-learn, Ollama

- Built an end-to-end machine learning system that predicts student academic risk using the Open University Learning Analytics Dataset (OULAD)Defined target users including average PC users seeking simplified file organization workflow

- Developed a Random Forest classification model to generate risk probabilities and student-level predictions
- Designed and deployed an interactive Streamlit web application for dataset upload, visualization, and real-time inference
- Integrated a local LLM-powered AI assistant (Nova) using Ollama (Llama 3) for natural language queries and explanation of model outputs
- Implemented automated intervention suggestions to support at-risk students based on model predictions

Personal Portfolio Website |Live Site (Vercel)| GitHub *(braxtonvogel)* HTML, CSS, TypeScript, Next.js

- Designed and developed a responsive personal portfolio website to showcase software engineering projects and technical skills
- Built interactive project display system with dynamic routing and structured project metadata for scalable expansion
- Implemented custom UI components including navigation, animated elements, and an AI-style skill matching widget for engagement
- Integrated GitHub + Vercel CI/CD pipeline enabling automatic deployments on every commit
- Optimized user experience with clean visual hierarchy, responsive layout, and mobile-first design principles

SKILLS

Technical:

- Python, Java, HTML, TypeScript, Rust
- Machine Learning: Scikit-learn, Random Forest, Data Analysis, Feature Engineering, Data Preprocessing
- AI / LLMs: Ollama, Llama 3, Prompt Engineering, Multi-Provider LLM Integration, OpenAI API, Anthropic API, Groq, Gemini, Cerebras
- Web Development: Streamlit, Spring Boot, Thymeleaf
- Software Engineering: Client-Server Architecture, Basic Networking Concepts, System Design Fundamentals
- Tools & Development: Git, GitHub, Notepad++
- Debugging & Problem Solving: Code Debugging, Error Resolution, Testing and Troubleshooting
- Computer Fundamentals: Hardware Fundamentals
- Security & Auth: JWT Authentication, bcrypt, Rate Limiting, Encrypted Key Storage
- Cloud & Infrastructure: Upstash Redis, QStash, Cloud Job Pipelines, Vercel Deployment

Interpersonal

- Teamwork & Collaboration
- Clear Technical Communication
- Critical Thinking
- Creativity, Problem Solving, and Innovation
- AI-Supported Research & Documentation

Professional:

- AI-Assisted Workflow Optimization
- Data-Driven Decision Making
- Operational Efficiency Management
- Process Improvement and Automation
- Report Generation & Report Writing

LEADERSHIP & EXTRACURRICULAR ACTIVITIES

- **Community Volunteer –** Local church service supporting food distribution for underdeveloped communities
- **Volunteer Participant –** Jeep Club led nonprofit fundraisers (Noah's ark foundation events, car shows, raffles)
- **Volunteer Support –** Veteran focused initiatives including Wreaths across America, flag placement, & fundraising